Summations solutions

A-E01 Arithmetic and square sums with shifted starts

Derivation and check

The first sum has six terms, with first term 10 and last term 25. Pairing gives $6(10+25)/2 = 105$. Subtracting prefixes gives the same result:

$$\sum_{k=4}^{9}(3k-2) = 3(45-6) - 2\cdot 6 = 105.$$

The second sum is $4+9+16+25 = 54$. The square-sum formula gives $5\cdot 6\cdot 11/6 - 1 = 54$.

To derive that formula, sum the cubic differences over $k = 0, \dots, n$:

$$\begin{aligned}\sum_{k=0}^{n}\left((k+1)^3 - k^3\right) &= (n+1)^3 \\ &= 3\sum_{k=0}^{n} k^2 + \frac{3n(n+1)}{2} + (n+1).\end{aligned}$$

Solving gives $\sum_{k=0}^{n} k^2 = n(n+1)(2n+1)/6$. The shifted starts subtract prefixes through $k=3$ and $k=1$, respectively. Replacing the upper limit by the number of terms would not make those adjustments.

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A-E02 Finite geometric sums and error

Derivation and check

For $x = -1/3$,

$$G_5 = \frac{1-(-1/3)^6}{1+1/3} = \left(1 - \frac{1}{729}\right)\frac{3}{4} = \frac{182}{243}.$$

The infinite sum is $3/4$, and its tail is

$$\frac{3}{4} - \frac{182}{243} = \frac{1}{972} = \frac{x^6}{1-x} > 0.$$

Thus the finite sum is slightly smaller than $3/4$. At $x = 1$, partial sums equal $n+1$. At $x = -1$, they oscillate between 0 and 1.

Let $W_n = \sum_{k=0}^{n} kx^k$. Then

$$W_n - xW_n = x + \cdots + x^n - nx^{n+1}.$$

Substituting the finite geometric sum and rearranging gives

$$W_n = \frac{x - (n+1)x^{n+1} + nx^{n+2}}{(1-x)^2}, \qquad x \neq 1.$$

Summations solutions (continued)

Only $|x| < 1$ permits $n|x|^n \to 0$ and the limit $x/(1-x)^2$. Endpoints require separate treatment.

A-E03 Telescoping products and logarithm premises

Derivation and check

Decompose the summand:

$$\frac{1}{(k+1)(k+3)} = \frac{1}{2}\left(\frac{1}{k+1} - \frac{1}{k+3}\right).$$

The telescoping sum is

$$\sum_{k=1}^{n} \frac{1}{(k+1)(k+3)} = \frac{1}{2}\left(\frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}\right).$$

At $n = 3$, $1/8 + 1/15 + 1/24 = 7/30$. Factor the product into two telescoping products:

$$\prod_{k=2}^{n}\left(1 - \frac{1}{k^2}\right) = \left(\prod_{k=2}^{n}\frac{k-1}{k}\right)\left(\prod_{k=2}^{n}\frac{k+1}{k}\right) = \frac{n+1}{2n}.$$

At $n = 3$, $(3/4)(8/9) = 2/3$. The logarithm identity

$$\ln\left(\prod_{k=1}^{n} a_k\right) = \sum_{k=1}^{n} \ln a_k$$

requires every factor to be a positive real number. Zero and negative factors have undefined real logarithms. A positive final product does not repair negative individual factors.

A-E04 Induction with fixed constants

Derivation and check

The base case is $S_0 = 1 \le c$, requiring $c \ge 1$. The induction step yields

$$S_{n+1} \le c5^n + 5^{n+1} = \left(\frac{c}{5} + 1\right)5^{n+1}.$$

It closes for $c \ge 5/4$. The fixed choice $c = 5/4$ works throughout. The last term gives $S_n \ge 5^n$, proving $S_n = \Theta(5^n)$.

For the arithmetic sum, assuming $\sum_{k=1}^{n} k \le cn$ would require

$$cn + n + 1 \le c(n+1), \qquad \text{or} \qquad n + 1 \le c.$$

That is impossible for all n with a fixed c. The error is the loss of a common constant, rather than the use of O notation itself.

Summations solutions (continued)

A-E05 Ratios and prefixes

Derivation and check

Let $a_k = k^2/4^k$. For $k \geq 2$,

$$\frac{a_{k+1}}{a_k} = \frac{1}{4}\left(1 + \frac{1}{k}\right)^2 \leq \frac{9}{16}.$$

The prefix is $a_1 = 1/4$, and the tail starts at $a_2 = 1/4$. Thus

$$\sum_{k=2}^{\infty} a_k \leq \frac{1/4}{1 - 9/16} = \frac{4}{7}, \qquad \sum_{k=1}^{\infty} a_k \leq \frac{1}{4} + \frac{4}{7} = \frac{23}{28}.$$

The exact weighted geometric sum at $x = 1/4$ is

$$\frac{x(1+x)}{(1-x)^3} = \frac{(1/4)(5/4)}{(3/4)^3} = \frac{20}{27} \approx 0.740741 < \frac{23}{28} \approx 0.821429.$$

Harmonic ratios $k/(k+1) < 1$ approach 1, and the series diverges. A uniform $\rho < 1$ is necessary for this geometric-tail argument.

A-E06 Two harmonic brackets

Derivation and check

The blocks for $n = 12$ are $[1]$, $[2, 3]$, $[4, 5, 6, 7]$, and $[8, 9, 10, 11, 12]$. Their sums are

$$1, \qquad \frac{5}{6}, \qquad \frac{319}{420}, \qquad \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12}.$$

The final block has five terms. Three full blocks give $H_{12} \geq 3/2$. Completing all four blocks gives $H_{12} \leq 4$. In general, for $n \geq 2$,

$$\frac{\lfloor \log_2 n \rfloor}{2} \leq H_n \leq \lfloor \log_2 n \rfloor + 1.$$

On $[k, k+1]$, $1/x \leq 1/k$. Sum the corresponding integrals to obtain $\int_1^{n+1} dx/x \leq H_n$. For $k \geq 2$, $1/x \geq 1/k$ on $[k-1, k]$, so $H_n - 1 \leq \int_1^n dx/x$. Therefore

$$\ln(n+1) \leq H_n \leq 1 + \ln n.$$

Integrating from zero encounters a nonintegrable singularity and gives an unhelpful infinite upper bound. Separating the first term is essential.

A-E07 p-series and changing parameters

Derivation and check

For a fixed $p > 1$, nonnegative partial sums increase and satisfy

Summations solutions (continued)

$$\sum_{k=1}^{n} \frac{1}{k^p} \le 1 + \int_1^{\infty} x^{-p}\,dx = 1 + \frac{1}{p-1}.$$

They are bounded, so they converge. At $p = 3$, the explicit upper bound is $3/2$. For $0 < p < 1$, integral bounds give

$$\frac{(n+1)^{1-p} - 1}{1-p} \le \sum_{k=1}^{n} \frac{1}{k^p} \le 1 + \frac{n^{1-p} - 1}{1-p},$$

proving $\Theta(n^{1-p})$. At $p = 1 + 1/n$, the earlier bound becomes $n + 1$, which is not $O(1)$. The theorem fixes p before the truncation limit grows. Changing p along a diagonal family does not preserve the constants.

A-E08 Nested loops and index sets

Derivation and check

Each j occurs for $i = 1, \ldots, j$, or j times. Finite reordering gives

$$\begin{aligned}
\sum_{i=1}^{n}\sum_{j=i}^{n}(2j+1) &= \sum_{j=1}^{n} j(2j+1) \\
&= 2\sum_{j=1}^{n} j^2 + \sum_{j=1}^{n} j \\
&= \frac{n(n+1)(2n+1)}{3} + \frac{n(n+1)}{2} \\
&= \frac{n(n+1)(4n+5)}{6}.
\end{aligned}$$

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At $n = 3$, this is $3 \cdot 4 \cdot 17/6 = 34$. Direct row sums are $15, 12, 7$, also totaling 34.

The upper and lower constants hidden in $\Theta(j)$ must be common to every legal (i, j) and all sufficiently large problem sizes. Unbounded coefficients that depend on position can change the total order. Finite reordering requires no infinite-convergence assumption.

Summations solutions (continued)

Verification code appendix: summation_checks.py

```
"""Finite rational identities and numerical integral-bound checks."""
from fractions import Fraction as F
from math import log,prod
if __name__=='__main__':
 for n in range(101):
  assert sum(k*k for k in range(n+1))==n*(n+1)*(2*n+1)//6
  assert sum(k**3 for k in range(n+1))==(n*(n+1)//2)**2
  assert sum(2*j+1 for i in range(1,n+1) for j in range(i,n+1))==n*(n+1)*(4*n+5)//6
  if n:
   h=sum((F(1,k) for k in range(1,n+1)),F(0));assert log(n+1)<=float(h)<=1+log(n)
   assert sum((F(1,k*(k+2)) for k in range(1,n+1)),F(0))==F(1,2)*(F(3,2)-F(1,n+1)-F(1,n+2))
   assert prod((1-F(1,k*k) for k in range(2,n+1)),start=F(1))==F(n+1,2*n)
  for x in [F(-1,3),F(1,2),F(2),F(-2)]:
   assert sum((x**k for k in range(n+1)),F(0))==(x**(n+1)-1)/(x-1)
   assert sum((k*x**k for k in range(n+1)),F(0))==(x-(n+1)*x**(n+1)+n*x**(n+2))/(1-x)**2
 assert sum((F(-1,3)**k for k in range(6)),F(0))==F(182,243)
 assert F(3,4)-F(182,243)==F(1,972)
 assert sum((F(1,(k+1)*(k+3)) for k in range(1,4)),F(0))==F(7,30)
 print('PASS finite n0..100 rational sums/products, weighted geometry, changed-index loops,
harmonic bounds')
```

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study